

Department of Mining Engineering
Indian Institute of Technology (ISM), Dhanbad

Examination: **MID SEMESTER**

B. Tech (MNE) 4th Semester

Session: **2023-24**

Subject: **MNC 205 ROCK MECHANICS**

Semester: **WINTER**

Instructions: 1. Answer **ANY TWO** questions

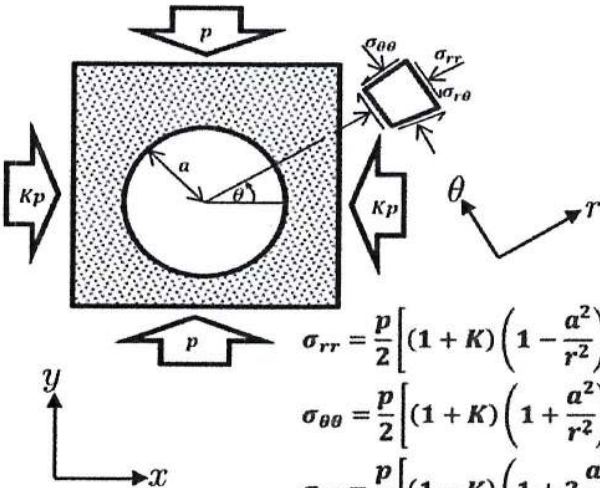
Time: **2 HOURS**

2. Neat and accurate sketches will assist you in solving the problems logically and faster

3. Assume logical data, where necessary

Max. Marks: **30**

4. Use the sign conventions applicable for rock mechanics.

Q. No.	Question	Marks
1	(i) Draw neat sketches and discuss the Cauchy's First and Second law of stress. 6 (ii) Draw the Mohr's Circle for (a) a rock core specimen loaded in uni-axial compression 1 (b) a thin square rock slab loaded in tension from two of its side. 1 (c) A square rock slab loaded hydro-statically 1 (iii) Kirsch in 1898, considered a circular opening in an infinite medium subjected to horizontal stress of $K \times p$ and a vertical stress of p. These stresses are known as the boundary stress or in situ stress. He gave the solution for finding out the components of stress tensor (σ_{rr}, $\sigma_{\theta\theta}$ and $\sigma_{r\theta}$) around a circular opening in the polar co-ordinate system with its origin at the center of the circular excavation, shown in the figure below. (a) Using the formula given below, determine the components of the stress tensor at point 'M' in the direction of 45° to the horizontal at an infinite distance. 1 (b) At the said point 'M', determine the components of the stress tensor in x-y coordinate system. 4 (c) Explain the inferences you draw from the results of point (a) and point (b) 1  $\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 - 4 \frac{a^2}{r^2} + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right]$ $\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right]$ $\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + 2 \frac{a^2}{r^2} - 3 \frac{a^4}{r^4} \right) \sin 2\theta \right]$	
2	(i) The rectangular state of stress at a point 'P' in a rock is given by the array $\begin{bmatrix} 5 & 5 & 3 \\ 5 & -5 & 4 \\ 3 & 4 & 8 \end{bmatrix}$ Determine the magnitude of the normal stress and the shear stress on a given plane. The direction cosines of the unit normal to the given plane are $n_x = \sqrt{\frac{1}{3}}$ and $n_y = \sqrt{\frac{1}{2}}$.	5

	(ii) The displacement field in rock mass is given by $\vec{u} = [87.5xz\hat{i} + 5yz^2\hat{j} + 2.5x^2y\hat{k}] \times 10^{-3}mm.$ Determine the normal strain at a point (1, 5, -3) along a line AB, whose direction cosines $n_x = n_y = n_z = \frac{1}{\sqrt{3}}$ (iii) The rectangular state of stress at a point 'P' in a rock is given by the array $\begin{bmatrix} 15 & 7 & 3 \\ 7 & 25 & 2 \\ 3 & 2 & 32 \end{bmatrix}$ Determine the unit vector along the intermediate principal plane.	5 5
Q. No.	Question	Marks
3	(a) In case of infinitesimal strains in two dimensions. i. Define the normal and shear strains. ii. Using a neat and labeled sketch, show that the normal strain in the x-direction is given by $\epsilon_{xx} = \frac{\partial}{\partial x}u(x, y)$ and the shear strain is given by $\epsilon_{xy} = \frac{1}{2} \left\{ \frac{\partial}{\partial y}u(x, y) + \frac{\partial}{\partial x}v(x, y) \right\}$. (b) The components of the state of stress at a point is given by the matrix $\begin{bmatrix} \sigma_x & 2 & 3 \\ 2 & 5 & 1.5 \\ 3 & 1.5 & -4 \end{bmatrix}$ Find the value of σ_x for which there exists a traction free plane and compute the direction cosines normal to the plane.	2+8 5

The relevant formula for solving the problem is given as under.
(The symbols have their usual meaning)

$$\tau' = L\tau L^T, \quad \vec{p} = [\tau] [n], \quad \sigma = \vec{p} \cdot \hat{n}$$

$$L = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \text{ for 2D}$$

$$L = \begin{bmatrix} \cos(\angle x'ox) & \cos(\angle x'oy) & \cos(\angle x'oz) \\ \cos(\angle y'ox) & \cos(\angle y'oy) & \cos(\angle y'oz) \\ \cos(\angle z'ox) & \cos(\angle z'oy) & \cos(\angle z'oz) \end{bmatrix} \text{ for 3D}$$



INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES)

DHANBAD - 826004, (JHARKHAND), INDIA

मध्यावदी परीक्षा (Mid Semester Examination)

①

Name : _____ Adm. No. : _____

Subject : _____ Part : _____ Semester : _____

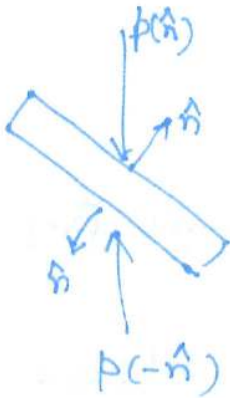
Signature of Invigilator : _____ Date : _____

Question No.	1	2	3	4	5	6	7	8	9	10	TOTAL
Marks Obtained											

Total Marks Obtained : _____ Signature of Examiner : _____

Ans-1(a): Cauchy's First Law of Stress

Consider a thin penny-shaped slab of rock having thickness 'h' and radius 'r'.



If the outward unit normal vector on the right face of the slab is denoted by $\hat{n}$

Then the outward unit normal vector on the left face of the slab is $-\hat{n}$.

$\therefore$ The total force acting on the face with outward unit normal vector $\hat{n}$ is $= \pi r^2 p(\hat{n})$.

Similarly on the opposite face it is $= \pi r^2 p(-\hat{n})$

The total force acting on the outer rim of the penny shaped slab will be given by the integral of the traction over the outer area and will be proportional to $2\pi r h$ (Surface area of the rim).

Performing the force balance on the slab yields

$$\pi r^2 p(\hat{n}) + \pi r^2 p(-\hat{n}) + 2\pi r h \vec{t} = 0$$

Where $\vec{t}$ is the mean traction over the outer rim.

If the thickness h of the slab is allowed to vanish, the third term will vanish, so the condition for equilibrium will be

$$\pi r^2 p(\hat{n}) + \pi r^2 p(-\hat{n}) = 0$$

$$\Rightarrow \boxed{\vec{p}(-\hat{n}) = -p(\hat{n})} \quad \text{--- (1)}$$

The equation (1) is known as "Cauchy's first law of stress".

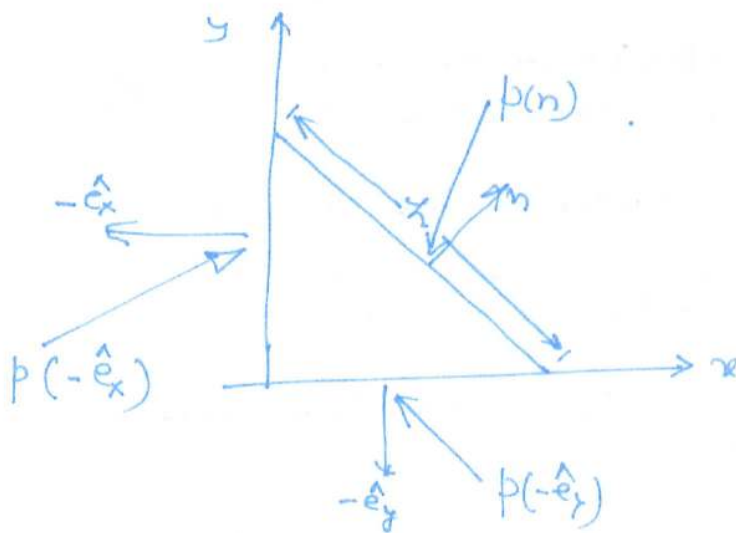
This law is an analogue of Newton's 3rd Law of motion.

"If the material to the left of a given plane exerts a traction $\vec{p}$ on the material on the right then the material on the right will exert a traction $-\vec{p}$ on the material to the left"

Cauchy's 2nd Law of Stress

3

Consider a triangular slab of rod as shown in the figure below.



Let the width of the slab in the z -direction $= w$.

Two of the faces of this slab have outward unit normal vector that coincides with the negative x & y coordinate directions respectively.

The third face has an outward unit normal $\hat{n} = \langle n_x, n_y \rangle$.

Let the length of the face with outward normal $\hat{n}$ $= h$.

The lengthy face that has
outward unit normal $-\hat{e}_x$ } = $h \hat{n}_x$

& its area = $h \hat{n}_x w$.

also the traction vector on this
face is denoted by } $p(-\hat{e}_x)$.

∴ The total force acting on this face is

$$= h n_x w \vec{p}(-\hat{e}_x).$$

Similarly for the face with unit normal
vector $-\hat{e}_y$

The total force acting on the
solid face } = $h w \hat{n}_y p(\hat{e}_y)$.

Now, the Force Balance on this slab leads to

$$h w n_x p(-\hat{e}_x) + h w \hat{n}_y \vec{p}(-\hat{e}_y) \\ + h w \vec{p}(\hat{n}) = 0$$

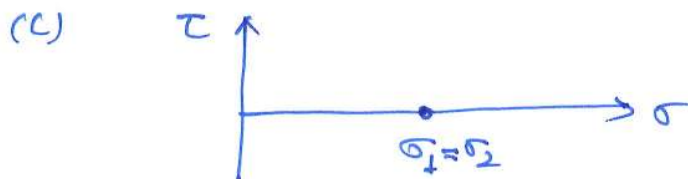
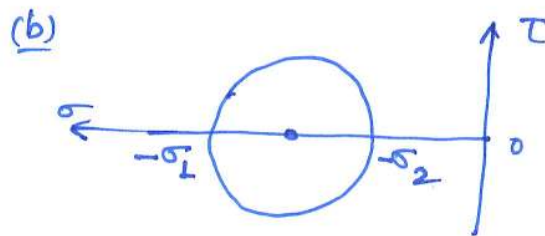
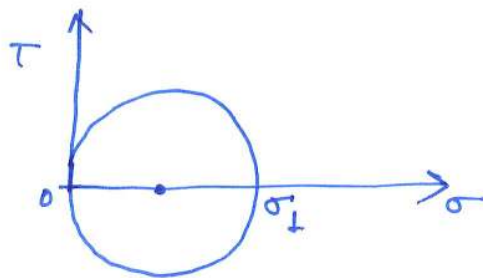
Cancelling out the common terms and
using the Cauchy's 1st Law we
have

$$\boxed{\vec{p}(\hat{n}) = n_x p(\hat{e}_x) + n_y p(\hat{e}_y)}.$$

Que: Draw the Mohr's Circle for

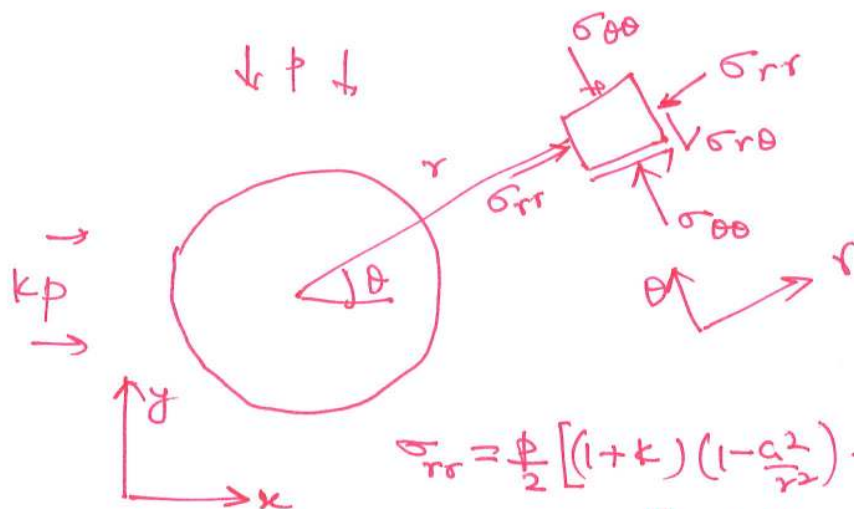
- Rock core specimen loaded in uniaxial compression.
- A thin square rock slab loaded in tension from two of its side
- A square rock slab loaded hydrostatically.

Ans- (a)



Ques: Kirsch in 1898, considered a circular opening in an infinite medium subjected to horizontal stress of Kp and a vertical stress of p' . These stresses are known as the boundary stress or in-situ stress.

He gave the solution for finding out the components of stress tensor (σ_{rr} , $\sigma_{\theta\theta}$ & $\sigma_{r\theta}$) around a circular opening in the polar co-ordinate system with its origin at the center of the circular excavation, shown in the figure below.



$$\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 + 4\frac{a^2}{r^2} + 3\frac{a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + 3\frac{a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + \frac{2a^2}{r^2} - \frac{3a^4}{r^4} \right) \sin 2\theta \right]$$

- using the given formula, determine the stress tensor at point 'M' in the direction of 45° to the horizontal at an infinite distance.
- At the said point 'M' determine the components of the stress tensor in any co-ordinate system.
- Explain the inferences you draw from the results of point (a) and point (b).

(1)

(a) Given $r = \infty$ & $\theta = 45^\circ$

$$\text{So, } \sigma_{rr} = \frac{p}{2}(1+k)$$

$$\sigma_{\theta\theta} = \frac{p}{2}(1+k)$$

$$\sigma_{r\theta} = \frac{p}{2}(1-k)$$

$$\therefore \tau = \frac{p}{2} \begin{bmatrix} 1+k & 1-k \\ 1-k & 1+k \end{bmatrix}$$

(b) For. Calculating the stress tensor in the xy -coordinate system, we have.

$$\theta = -45^\circ$$

$$\therefore L = \begin{bmatrix} \cos(-45^\circ) & \sin(-45^\circ) \\ -\sin(-45^\circ) & \cos(-45^\circ) \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\bullet \quad L^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\text{Now } \tau' = L \tau L^T$$

$$\begin{aligned} \Rightarrow \tau' &= \frac{p}{2} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1+k & 1-k \\ 1-k & 1+k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \\ &= \frac{p}{4} \begin{bmatrix} 1+k-1+k & 1-k-1-k \\ 1+k+1-k & 1-k+1+k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \end{aligned}$$

(2)

$$\begin{aligned}\tau' &= \frac{p}{4} \begin{bmatrix} 2k & -2k \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \\ &= \frac{p}{4} \begin{bmatrix} 4k & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} pk & 0 \\ 0 & p \end{bmatrix}\end{aligned}$$

(c)

- * the stress^{tensor} at point 'M' in the ro Co-ordinate System is not a principal stress system.
- * However, in the xy -Co-ordinate System the stress tensor is a principal stress system & is equal to the in-situ stress.
- * Therefore, the influence of the excavation is nil at an infinite distance.

2(i). The Rectangular State of stress at a point 'P' in a rock is given by the array

$$\begin{bmatrix} 5 & 5 & 3 \\ 5 & -5 & 4 \\ 3 & 4 & 8 \end{bmatrix}$$

Determine the magnitude of the normal stress and the shear stress on a given plane.

The direction cosines of the unit normal to the given plane are

$$n_x = \frac{1}{\sqrt{3}} \quad \text{e} \quad n_y = \frac{1}{\sqrt{2}}$$

Solution: Given $n_x = \frac{1}{\sqrt{3}}$, $n_y = \frac{1}{\sqrt{2}}$

$$\text{But } \left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + n_z^2 = 1$$

$$\Rightarrow n_z^2 = 1 - \frac{1}{3} - \frac{1}{2} = \frac{6-2-3}{6} = \frac{1}{6}$$

$$\Rightarrow n_z = \frac{1}{\sqrt{6}}$$

$$\begin{aligned} \vec{P} = [\tau][h] &= \begin{bmatrix} 5 & 5 & 3 \\ 5 & -5 & 4 \\ 3 & 4 & 8 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} \end{bmatrix} \\ &= \begin{bmatrix} 7.6470 \\ 0.9842 \\ 7.8265 \end{bmatrix} \end{aligned}$$

$$\sigma = \vec{P} \cdot \hat{n} = [7.6470 \quad 0.9842 \quad 7.8265] \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$

$$\sigma = 8.306$$

$$\Rightarrow \tau = \sqrt{|\vec{P}|^2 - \sigma^2} = 7.190 \quad \underline{\text{Ans}}$$

Ques: The displacement field in rock mass is given by

$$\vec{u} = [87.5xz \hat{i} + 5yz^2 \hat{j} + 2.5x^2y \hat{k}] \times 10^{-3} \text{ mm.}$$

Determine the normal strain at point $(1, 5, -3)$ along a line AB, whose dds are $n_x = n_y = n_z = \frac{1}{\sqrt{3}}$.

Solution:

$$\nabla \vec{u} = \begin{bmatrix} 87.5z & 0 & 87.5x \\ 0 & 5z^2 & 10yz \\ 5xy & 2.5x^2 & 0 \end{bmatrix} \times 10^{-3}.$$

$$(\nabla \vec{u})_{(1,5,-3)} = \begin{bmatrix} 87.5 \times (-3) & 0 & 87.5 \times 1 \\ 0 & 5 \times (-3)^2 & 10 \times 5 \times (-3) \\ 5 \times 1 \times 5 & 2.5 \times 1^2 & 0 \end{bmatrix} \times 10^{-3}$$

$$= \begin{bmatrix} -262.5 & 0 & 87.5 \\ 0 & 45 & -150 \\ 25 & 2.5 & 0 \end{bmatrix} \times 10^{-3}.$$

$$\nabla \vec{u} + (\nabla \vec{u})^T = \begin{bmatrix} -262.5 & 0 & 87.5 \\ 0 & 45 & -150 \\ 25 & 2.5 & 0 \end{bmatrix} \times 10^{-3} + \begin{bmatrix} -262.5 & 0 & 25 \\ 0 & 45 & 2.5 \\ 87.5 & -150 & 0 \end{bmatrix} \times 10^{-3}$$

$$= \begin{bmatrix} -2 \times 262.5 & 0 & \frac{225}{2} \\ 0 & 90 & -\frac{295}{2} \\ \frac{225}{2} & -\frac{295}{2} & 0 \end{bmatrix} \times 10^{-3}.$$

$$\epsilon = \frac{1}{2} [\nabla \vec{u} + (\nabla \vec{u})^T] = \frac{10^{-3}}{2} \begin{bmatrix} -2 \times 262.5 & 0 & \frac{225}{2} \\ 0 & 90 & -\frac{295}{2} \\ \frac{225}{2} & -\frac{295}{2} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -262.5 & 0 & 56.25 \\ 0 & 45 & -73.75 \\ 56.25 & -73.75 & 0 \end{bmatrix} \times 10^{-3}.$$

$$[\epsilon][\hat{n}] = \begin{bmatrix} -119.8785 \\ -16.5988 \\ -10.1036 \end{bmatrix} \times 10^{-3}$$

$$\epsilon_{AB} = \{ [\epsilon][\hat{n}] \}^T \cdot [\hat{n}]$$

$$= -84.167 \times 10^{-3}$$

Ques: The rectangular state of stress at point 'P' in a rock is given by the array

$$\begin{bmatrix} 15 & 7 & 3 \\ 7 & 25 & 2 \\ 3 & 2 & 32 \end{bmatrix}$$

Determine the unit vector along the intermediate principal value.

Soln:-

$$P = \text{Trace}(\tau) = 15 + 25 + 32 = 72$$

$Q =$ Sum of the minors of the diagonal terms

$$= \begin{vmatrix} 25 & 2 \\ 2 & 32 \end{vmatrix} + \begin{vmatrix} 15 & 3 \\ 3 & 32 \end{vmatrix} + \begin{vmatrix} 15 & 7 \\ 7 & 25 \end{vmatrix}$$
$$= 796 + 471 + 326 = 1593$$

$$R = \begin{vmatrix} 15 & 7 & 3 \\ 7 & 25 & 2 \\ 3 & 2 & 32 \end{vmatrix} = 10231$$

$$\lambda^3 - P\lambda^2 + Q\lambda - R = 0$$

$$\lambda^3 - 72\lambda^2 + 1593\lambda - 10231 = 0$$

$$\lambda_1 = 11.246$$

$$\lambda_2 = 33.982$$

$$\lambda_3 = 26.773$$

The intermediate principal stress is $\lambda_3 = 26.773$

Let the vector associated with λ_2 be n_x, n_y, n_z

$$\begin{bmatrix} 15-26.773 & 7 & 3 \\ 7 & 25-26.773 & 2 \\ 3 & 2 & 32-26.773 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -11.773 & 7 & 3 \\ 7 & -1.773 & 2 \\ 3 & 2 & 5.227 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left(\begin{array}{ccc|c} 1 & -0.595 & -0.255 & 0 \\ 1 & -0.253 & 0.286 & 0 \\ 1 & 0.667 & 1.742 & 0 \end{array} \right)$$

$$R_1 \rightarrow R_1 / (-11.773)$$

$$R_2 \rightarrow R_2 / 7$$

$$R_3 \rightarrow R_3 / 3$$

$$\left(\begin{array}{ccc|c} 1 & -0.595 & -0.255 & 0 \\ 1 & -0.253 & 0.286 & 0 \\ 0 & 0.920 & 1.456 & 0 \end{array} \right) \quad R_3 \rightarrow R_3 - R_2$$

$$\left(\begin{array}{ccc|c} 1 & -0.595 & -0.255 & 0 \\ 0 & 0.342 & 0.541 & 0 \\ 0 & 0.920 & 1.456 & 0 \end{array} \right) \quad R_2 \rightarrow R_2 - R_1$$

$$\left(\begin{array}{ccc|c} 1 & -0.595 & -0.255 & 0 \\ 0 & 1 & 1.582 & 0 \\ 0 & 1 & 1.583 & 0 \end{array} \right) \quad \begin{array}{l} R_2 \rightarrow R_2 / 0.341 \\ R_3 \rightarrow R_3 / 0.920 \end{array}$$

$$\text{let } n_z = 1$$

$$\Rightarrow n_y = -1.583$$

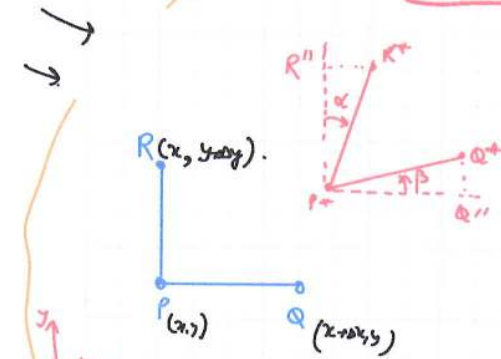
$$n_x - 0.595 \times (-1.583) - 0.255 \times 1 = 0$$

$$n_x = -0.687$$

Normalizing

$$\left. \begin{array}{l} n_x = -0.344 \\ n_y = -0.794 \\ n_z = +0.501 \end{array} \right\} \underline{A_p}.$$

Infinitesimal Strain in Two Dimensions



$$P^* = P - u(P) = (x, y) - [u(x, y), v(x, y)]$$

$$= [x - u(x, y), y - v(x, y)] \quad \rightarrow \textcircled{1}$$

$$Q^* = Q - u(Q) = [x + \Delta x, y] - [u(x + \Delta x, y), v(x + \Delta x, y)]$$

$$= \{ [x + \Delta x - u(x + \Delta x, y)], [y - v(x + \Delta x, y)] \} \quad \rightarrow \textcircled{2}$$

$$R^* = R - u(R) = (x, y + \Delta y) - [u(x, y + \Delta y), v(x, y + \Delta y)]$$

$$= \{ [x - u(x, y + \Delta y)], [y + \Delta y - v(x, y + \Delta y)] \} \quad \rightarrow \textcircled{3}$$

2D-4D
u-strain.
2D-4D.

Here, Q'' = projecting line P^*Q^* on horizontal line passing through P^*
 R'' = proj. of line P^*R^* on vert. line passing through P^* .

Now, expressing each of the displacements appearing in P^* , Q^* & R^* as Taylor's series & ignore terms higher than first order.

$$P^* = (x - u, y - v) \quad \rightarrow \textcircled{4}$$

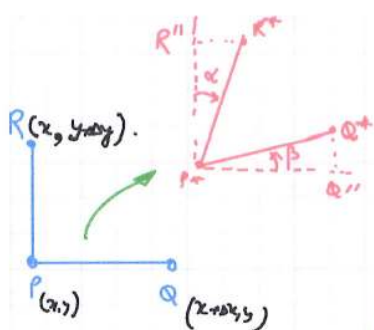
$$Q^* = \left[x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v - \frac{\partial v}{\partial x} \Delta x \right] \quad \rightarrow \textcircled{5}$$

Assuming that
PD are evaluated
at point (x, y).

$$R^* = \left[x - u - \frac{\partial u}{\partial y} \Delta y, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y \right] \quad \rightarrow \textcircled{6}$$

$$Q'' = \left[x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v \right] \quad \rightarrow \textcircled{7}$$

$$R'' = \left[x - u, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y \right] \quad \rightarrow \textcircled{8}$$



Normal Strain (ϵ_{xx}).

The normal strain ϵ_{xx} is defined as the fractional shortening of the line that is initially oriented along the x-axis.

ie: $\epsilon_{xx} = \text{Contractor undergone by } PQ \text{ in limit } \Delta x \rightarrow \underline{0}$

Initial length of $PQ = \Delta x$.

Final length of $PQ = P^*Q^*$.

$$\begin{aligned} |P^*Q^*|^2 &= |P^*Q'|^2 + |Q'^*Q''|^2 \\ &= \left(\Delta x - \frac{\partial u}{\partial x} \Delta x\right)^2 + \left(\frac{\partial v}{\partial x} \Delta x\right)^2 \quad \leftarrow \\ &= (\Delta x)^2 \left[1 - 2 \frac{\partial u}{\partial x} + \underbrace{\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2}_{\text{Ignore}} \right] \quad \text{--- 9} \end{aligned}$$

Ignoring Higher order terms.

& using the fact $(1 - 2\delta)^{1/2} = 1 - \delta \quad \forall \delta \ll 1$

$$|P^*Q^*|^2 = (\Delta x)^2 \left[1 - 2 \frac{\partial u}{\partial x} \right]$$

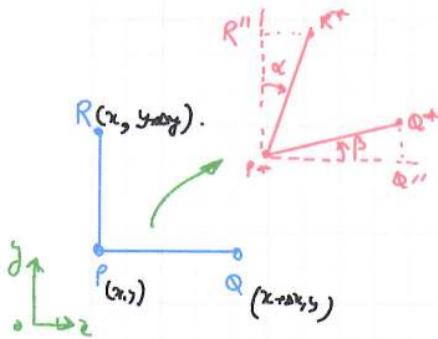
$$P^*Q^* = \Delta x \left(1 - \frac{\partial u}{\partial x} \right) \quad \text{--- 10}$$

Normal strain in x-direction at point (x, y)

$$\epsilon_{xx} = \lim_{\Delta x \rightarrow 0} \frac{PQ - P^*Q^*}{PQ} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x - \Delta x \left(1 - \frac{\partial u}{\partial x} \right)}{\Delta x} = \frac{\partial u}{\partial x}$$

$$\therefore \boxed{\epsilon_{xx} = \frac{\partial u}{\partial x}} \quad \parallel y \rightarrow \quad \boxed{\epsilon_{yy} = \frac{\partial v}{\partial y}}$$

Shear Strain in 2D.



Initially $P, Q \& R$ form a right angle.

After deformation $\rightarrow$ acute angle.

Change in angle is quantified as Shear Strain.

Convention $\rightarrow$
of the Res. Mech.

Defn:- The Shear Strain (ϵ_{xy}) is defined as half of the increase in angle formed by two infinitesimal line segments that are initially parallel to the x & y axes.

$$\epsilon_{xy} = \frac{1}{2} \lim_{\Delta x, \Delta y \rightarrow 0} (\angle R^* P^* Q^* - \angle RPQ)$$

$$\angle R^* P^* Q^* = 90^\circ - \alpha - \beta \quad \angle RPQ = 90^\circ$$

$$\tan \alpha = \left| \frac{R^* R''}{P^* R''} \right| = \frac{-\frac{\partial u}{\partial y} \Delta y}{\Delta y (1 - \frac{\partial v}{\partial y})}$$

For derivation see the Co-ordinating of $R^* R''$ & $P^* R''$ defined earlier.

$$\left(1 - \frac{\partial v}{\partial y}\right)^{-1} = 1 + \frac{\partial v}{\partial y} + \dots$$

$$= -\frac{\partial u}{\partial y} \left(1 + \frac{\partial v}{\partial y}\right)$$

$$= -\frac{\partial u}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial y}$$

(infinitesimal strain it is neglected.)

$$= -\frac{\partial u}{\partial y}$$

$$\boxed{\alpha = -\frac{\partial u}{\partial y}}$$

$$\parallel \rightarrow \boxed{\beta = -\frac{\partial v}{\partial x}}$$

$$\therefore \boxed{\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)}$$

The pattern shows that

$\epsilon_{xy} = \text{Avg of}$

- (i) P.D of displacement in x -dir w.r.t y
- (ii) P.D of displacement in y -dir w.r.t x .

By defn. if ϵ_{yx} is defined. then necessarily $= \epsilon_{xy}$.

$$\epsilon = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} \\ \epsilon_{xy} & \epsilon_{yy} \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{\partial v}{\partial y} \end{bmatrix}$$

Imp:- The engineering strain $\gamma_{xy} = 2\epsilon_{xy}$.

Some Properties of Strain Matrix.

- ① Strain matrix is equal to the symmetric part of displacement gradient $\nabla \vec{u}$.

$$\epsilon = \text{Sym}(\nabla \vec{u}) = \frac{1}{2} \{ \nabla \vec{u} + (\nabla \vec{u})^T \}.$$

$$\nabla \vec{u} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix}.$$

- ② The displacement gradient and hence also the Strain is a 2nd order tensor.

Under the rotation of Co-ordinate axes.
The Component transform according to the law

$$\epsilon' = L \epsilon L^T.$$

Proof:- Under the rotation of Co-ordinate axes.

Any Vector

$$\left. \begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= -x \sin \theta + y \cos \theta \end{aligned} \right\} \checkmark$$

$$\underline{[x']} = L [x]$$

$$L^T [x'] = [x].$$

Ques: The components of the state of stress at a point is given by the matrix .

$$\begin{bmatrix} \sigma_x & 2 & 3 \\ 2 & 5 & 1.5 \\ 3 & 1.5 & -4 \end{bmatrix}$$

Find the value of σ_x for which there exists a traction free plane and compute the direction co-sines of normal to the plane.

Soln:- Let (n_x, n_y, n_z) be the d.c's of the plane normal.

Then traction on the plane

$$\vec{P} = [\tau] [\hat{n}] = 0$$

$$\text{or, } \begin{bmatrix} \sigma_x & 2 & 3 \\ 2 & 5 & 1.5 \\ 3 & 1.5 & -4 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Since n_x, n_y, n_z are not all zero

$$\therefore \begin{vmatrix} \sigma_x & 2 & 3 \\ 2 & 5 & 1.5 \\ 3 & 1.5 & -4 \end{vmatrix} = 0$$

$$\sigma_x(-20-2 \cdot 25) - 2(-8-4 \cdot 5) + 3(3-15) = 0$$

$$-22 \cdot 25 \sigma_x + 25 - 36 = 0$$

$$\text{or. } \sigma_x = \frac{-11}{22 \cdot 25} = -0.494$$

$$\therefore \begin{bmatrix} -0.494 & 2 & 3 \\ 2 & 5 & 1.5 \\ 3 & 1.5 & -4 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving the homogeneous Equation by Gauss Elimination.

$$\left(\begin{array}{ccc|c} 1 & -4.049 & -6.073 & 0 \\ 2 & 2.5 & 0.75 & 0 \\ 3 & 0.5 & -1.333 & 0 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 / (-0.494) \\ R_2 \rightarrow R_2 / 2 \\ R_3 \rightarrow R_3 / 3 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & -4.049 & -6.073 & 0 \\ 0 & 6.549 & 6.823 & 0 \\ 0 & -2 & -2.083 & 0 \end{array} \right) \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_2 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & -4.049 & -6.073 & 0 \\ 0 & 1 & 1.042 & 0 \\ 0 & 1 & 1.042 & 0 \end{array} \right) \begin{array}{l} R_2 \rightarrow R_2 / 6.549 \\ R_3 \rightarrow R_3 / (-2) \end{array}$$

one of the Equations is redundant.

$$\text{let } n_z = 1$$

$$\Rightarrow n_y = -1.042$$

$$n_x - 4.049 \times (-1.042) - 6.073 = 0$$

$$\Rightarrow n_x = 1.854$$

Normalizing n_x , n_y & n_z

$$\left\{ \begin{array}{l} n_x = 0.789 \\ n_y = -0.442 \\ n_z = 0.426 \end{array} \right\} \underline{\text{Ans}} .$$